CSCI 3308 - Team 7

Task Tracker

Worksite task coordination for managers, workers, and contractors

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Members

- Hudson Siglinger (Platform and API)
- Winston Thompson (DB)
- Hudson Vo (UI)
- Ryken Armstrong (Middleware)
- Denzel Gurung (MiddleWare)
- Joshua Stone (UI)

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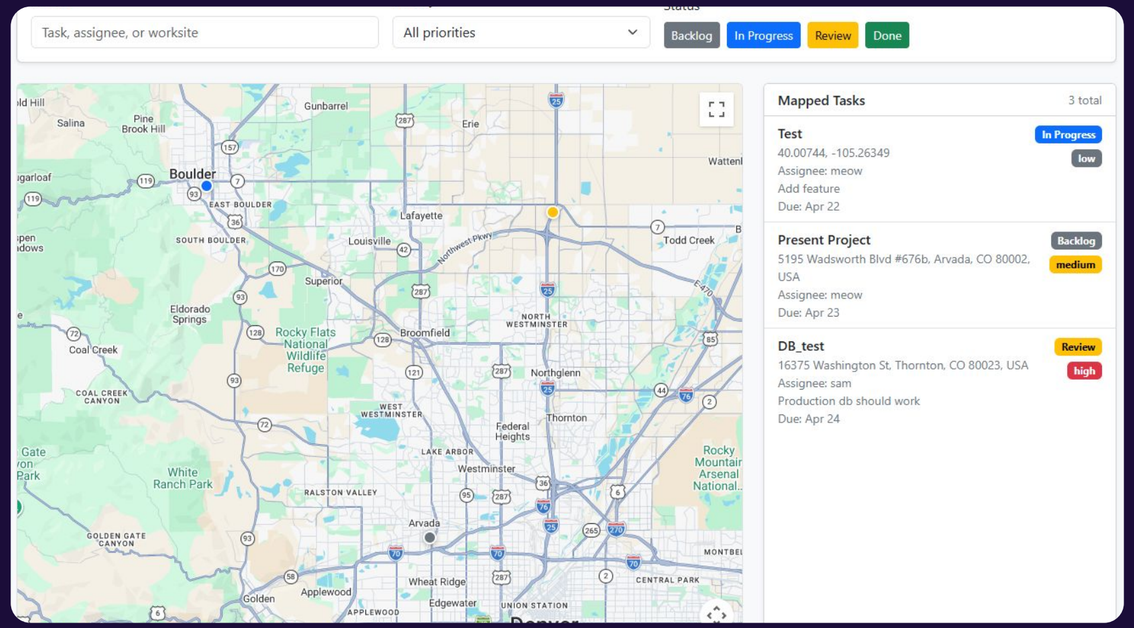
What Is Task Tracker?

Task Tracker is a web application for coordinating work-order tasks across real-world worksites.

Vision: To help teams communicate, plan, and track operational work from one secure and organized system.

- Centralizes work orders and task status
- Connects tasks to real-world worksites
- Supports managers, workers, and contractors with different access levels
- Improves visibility, planning, and accountability



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Tools Used – Development And Collaboration

Tool / Method - Rating	Purpose
GitHub/Git - 5/5	Version control system and shared repository
GitHub Projects / Project Tracker - 4/5	Built-in workflows for reviews, approvals, and handoffs
Visual Studio Code - 5/5	Main code editor and development environment
Discord/Zoom - 4/5	Remote meetings and team coordination

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Tools Used – Application Stack

Tool / Method - Rating	Purpose
Node.js - 5/5	Application server runtime
Express/PostgreSQL - 5/5	Backend web framework and API routes/Relational database
Bootstrap/HTML/CSS/JS - 3/5	Responsive frontend UI styling and browser behavior
Google Maps JavaScript API - 4/5	Worksite search, dropped pins, and map dashboard

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Tools Used – Testing, Deployment, And Support

Tool / Method - Rating	Purpose
Docker - 5/5	Local deployment environment for web and database containers
Mocha/Chai - 4/5	Test runner/Assertion library for tests
chai-http - 4/5	API endpoint testing
bcrypt - 5/5	Password hashing

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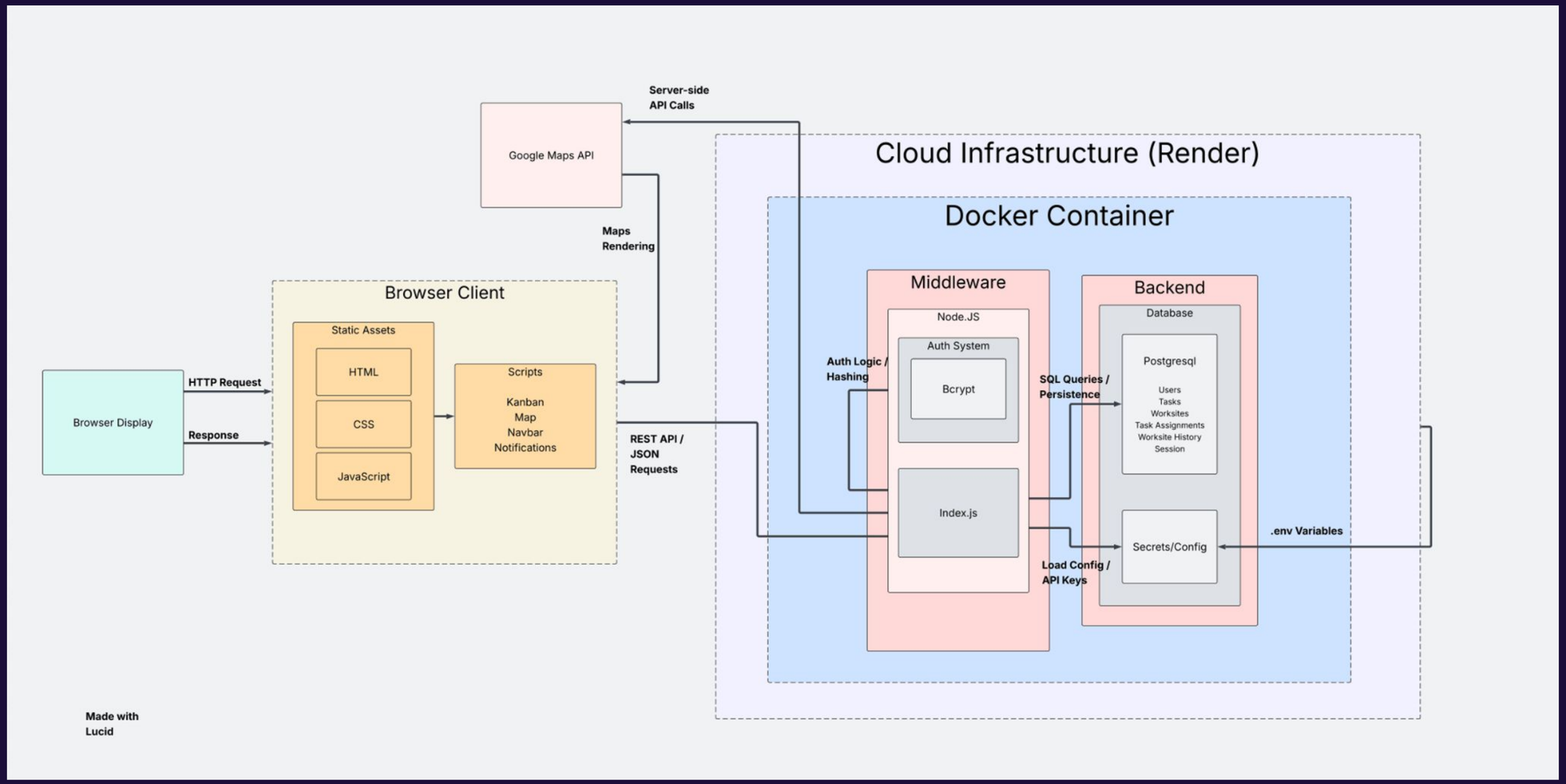
Development Methodologies

Agile / iterative process - 5/5
Task tracking - 4/5
Peer coordination - 4/5
Automated testing - 5/5

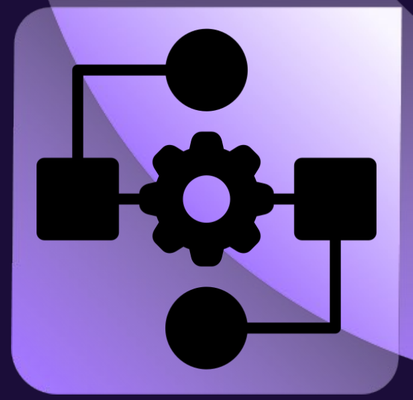
Agile	Iterative feature development, Let the team prioritize core features first
Project-board	Helped split a large project into manageable user stories
Peer coordination & Team check-ins	Made it easier to adjust scope as Maps, Docker, and role access became more complex
Testing	Manual quantified user acceptance testing and automated API tests kept testing turnaround minimal

Application Architecture



Data Flow

How Data Moves Through The App



#1

Manager logs in through the navbar modal

#2

Express verifies credentials and returns a JWT

#3

Manager creates a task on the Kanban board

#4

Browser sends task data and JWT to the Express API

#5

Express checks the manager/admin role

#6

Express writes task and worksite data to PostgreSQL

#7

Kanban board reloads updated task data

#8

Map dashboard reads mapped tasks and displays markers with Google Maps

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Challenges

Google Maps integration complexity

★★★

Added config/status checks and separated map logic into frontend scripts | Required extra debugging time

Role-based access control

★★

Added JWT role claims and backend route protection | Made auth more central to the design

Scope/Merge coordination

★★★★★

Focused on Kanban, auth, maps, and service status as core value | Some advanced ideas moved to future scope

Team waited on some commits and communicated before integration | Reduced conflicts but slowed some work

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Future Scope

- External notifications for task assignment, status changes, and overdue work
- Richer assignment workflow for teams, contractors, reviewers, and observers
- Stronger audit history showing who changed a task, when, and why.

- Navigation and planning assistance for contractors.
- More detailed manager analytics, such as worksite workload and completion trends
- ~~Production deployment with hosted database, HTTPS, and stronger secret management~~

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Please enjoy our live demo

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Questions?

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